Viva skill report · Student 01

Repo	student01/shop-api <https://github.com/student01/shop-api/tree/a1b2c3d>	Commit	a1b2c3d (main)
Cohort	CSC413 · Section B	Sat	10 Nov 2026, 10:05–10:47

! **Provisional.** AI scores are advisory until your examiner, Dr. Rao, confirms or overrides them.

Viva	Hidden tests	Viva time used
82/100	**4**/5	**11:04** of 12:00

Questions

Q1 src/db/pool.js · 3–12 <https://github.com/student01/shop-api/blob/a1b2c3d/src/db/pool.js#L3-L12>

[Strong]

Why does the pool set `connectionLimit: 10`, and what happens to request 11?

> *It queues until a connection is released; it doesn't fail unless the queue limit is hit.*

Why: Correct queueing behaviour, and named the queueLimit option without being asked.

Q2 src/routes/items.js · 18–27 <https://github.com/student01/shop-api/blob/a1b2c3d/src/routes/items.js#L18-L27>

[Strong]

This route builds SQL with ? placeholders. What attack does that stop?

> *SQL injection, because the driver sends the values separately from the query text.*

Why: Right attack and the right mechanism.

PROVISIONAL · NOT A FINAL GRADE

Q3 `src/routes/orders.js · 42–58` <https://github.com/student01/shop-api/blob/a1b2c3d/src/routes/orders.js#L42-L58>

[Partial]

You open a transaction on line 44. If the insert on line 51 throws, what happens to the stock count, and where is the connection released?

> *The UPDATE already ran, so stock drops but no order exists. The catch never calls rollback, and conn.release() is missing on both paths.*

Why: Found both bugs, but in the follow-up said the stock is restored when the connection closes, which isn't true while the transaction is still open.

Q4 `src/middleware/auth.js · 8–21` <https://github.com/student01/shop-api/blob/a1b2c3d/src/middleware/auth.js#L8-L21>

[Strong]

What does this middleware send back when the token has expired, and why that status code?

> *401, because the client isn't authenticated any more. 403 would mean "logged in but not allowed".*

Why: Correct code and a clear 401 vs 403 distinction.

Q5 `src/app.js · 30–36` <https://github.com/student01/shop-api/blob/a1b2c3d/src/app.js#L30-L36>

[Partial]

Your error handler is registered before the routes. Which errors will it never see?

> *Errors thrown inside the routes, because Express runs middleware in order.*

Why: Right idea, but didn't notice that async errors also need to be passed to next().

Q6 `src/services/cart.js · 40–52` <https://github.com/student01/shop-api/blob/a1b2c3d/src/services/cart.js#L40-L52>

[Missed]

Two requests add the last item to two carts at once. Which line lets both succeed?

> No answer within the time limit. Scored 0 for this question; the follow-up was skipped.

Why: The timer ran out. A good answer points to line 44, which reads stock before writing it without a lock.

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Q7 `package.json · 6–9` <https://github.com/student01/shop-api/blob/a1b2c3d/package.json#L6-L9>

[Strong]

What does `npm test` actually run here?

> *node --test on the test folder, the built-in runner, no Jest.*

Why: Exact.

Q8 `src/routes/orders.js · 60–74` <https://github.com/student01/shop-api/blob/a1b2c3d/src/routes/orders.js#L60-L74>

[Strong]

GET /orders has no LIMIT. What breaks first as the table grows?

> *Response time and memory, since every row is loaded and serialised. I'd add page and limit query params.*

Why: Identified the cost and proposed a fix.

Change request · hidden tests

Task: make a failed order leave stock unchanged and always release the connection.

TEST	RESULT	TIME
`rolls back stock when insert fails`	Pass	41 ms
`releases connection on success`	Pass	12 ms
`releases connection on error`	Pass	15 ms
`returns 409 when stock is short`	**Fail · got 500**	**9 ms**
Sandbox: no network, non-root, 30 s limit, destroyed after the run.		
`existing order routes still pass`	Pass	88 ms

Examiner signature and date

Student acknowledgement

Generated by Grill from commit a1b2c3d. Each citation above links to the exact lines on GitHub.